

SANTHOSH EKAMBARAM

Data Engineer | AWS | GCP | Big Data | ETL Pipelines

Dubai, UAE | santhosh02072k@gmail.com | 0503481037 | Residence Visa | Available Immediately
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PROFESSIONAL SUMMARY

M.S. Computer Science, Illinois Institute of Technology, Chicago. Results-driven Data Engineer with hands-on experience designing scalable ETL/ELT pipelines, data warehouses, and real-time data systems. Proficient in Python, PySpark, SQL, AWS, and GCP. Experienced in data ingestion, data integration, data modeling, multi-dimensional analysis, and delivering analytics-ready datasets for business intelligence. AWS Certified Solutions Architect. Based in Dubai on a Residence Visa — available immediately.

TECHNICAL SKILLS

Programming & Data: Python, SQL, Pandas, PySpark, Scala (basic)

Big Data & Processing: Apache Spark, Hadoop, Hive, Kafka, Flink, MapReduce, AWS Glue

Cloud Platforms: AWS (S3, Glue, Redshift, Lambda, Kinesis, Athena) | GCP (BigQuery, Cloud Storage, Dataflow)

ETL & Data Systems: ETL/ELT pipeline design, data ingestion, data integration, data warehousing

Data Modeling: Star schema, fact/dimension tables, schema design, multi-dimensional modeling

Streaming & Real-Time: Apache Kafka, Apache Flink, ClickHouse, AWS Kinesis

Analytics & BI: Power BI, Tableau, QuickSight, Matplotlib

Machine Learning: XGBoost, Logistic Regression, SMOTE, SHAP, Scikit-learn, TensorFlow, PyTorch

PROFESSIONAL EXPERIENCE

Data Engineering Intern | [Capgemini Invent, Dubai, UAE](#) | Jun 2024 – Aug 2024

- Designed and deployed an end-to-end ETL pipeline using Python and AWS (S3, Lambda) processing 500K+ financial records daily, enabling analytics-ready datasets for credit risk use cases.
- Performed data analysis, data ingestion, and integration from multiple financial data sources — ensuring accuracy and completeness of output datasets across 3 source systems.
- Built a credit risk defaulter detection model using XGBoost achieving 92% AUC score; addressed class imbalance with SMOTE and improved explainability using SHAP values.
- Collaborated with cross-functional teams on pipeline validation, automation, and real-time data delivery for financial risk analytics.
- Optimized SQL queries and data transformation logic, reducing pipeline execution time by 20%.

KEY PROJECTS

Real-Time User Engagement Analytics Pipeline (Kafka + Flink + ClickHouse) Apr 2025 – May 2025

- Designed and built a real-time streaming pipeline using Apache Kafka and Flink to process high-throughput event streams into analytics-ready datasets.
- Implemented event-time windowing and aggregations; stored results in ClickHouse with MergeTree schema for sub-second OLAP query performance on 1M+ events.
- Built SQL reporting layer for real-time KPI dashboards powering data-driven platform decisions.

Hive + Spark Batch Data Warehouse on Hadoop

May 2025 – Jun 2025

- Architected a Hadoop-based batch data warehouse using Hive and PySpark with partitioning and bucketing for enterprise-scale processing.
- Deployed Presto/Trino for fast interactive SQL querying; designed multi-dimensional data models for business reporting and growth analysis.

E-Commerce Sales Analytics Pipeline (AWS + ETL + SQL)

Jan 2025 – Feb 2025

- Built scalable ETL pipeline using AWS Glue and Redshift with star schema data model; enabled QuickSight dashboards for executive KPI reporting.

- Scheduled PySpark ETL jobs for daily data loads with partitioning for optimized query performance.

Google Cloud Data Analytics Pipeline (GCP | BigQuery | SQL)

Mar 2025 – Apr 2025

- Designed GCP analytics pipeline using Cloud Storage and BigQuery; wrote optimized SQL with partitioned tables for cost-efficient enterprise-scale reporting.

EDUCATION

M.S. in Computer Science — Illinois Institute of Technology, Chicago, USA *2023 – 2025*

B.Tech in Information Technology — Sri Sairam Engineering College, Tamil Nadu, India *2019 – 2022*

Diploma in Computer Engineering — Sairam Polytechnic College, Tamil Nadu, India *2016 – 2019*

CERTIFICATIONS

AWS Certified Solutions Architect — Amazon Web Services (2025)

IBM CI/CD Certification — IBM (2025)